Element I: Testing, Data Collection, and Analysis

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students’ ability to rest.

Our tests are designed to address our high-priority design requirements: allowing insects to be neutralized or caught in one area, being small (less than 14"x14"x14"), weighing less than one pound, and being assembled/applied within 10 minutes. To assess the effectiveness of our insect trap, we conducted tests and collected data related to these requirements. The data shows how well our solution meets the design requirements and the needs of Student E. This information will guide future design improvements.

The first prototype underwent testing to evaluate its effectiveness. The prototype was placed outside of Student E's room to observe its performance in a real-world setting and to determine if it would capture any insects. The testing revealed that the prototype captured 3 mosquitoes and some smaller insects. This result indicated that the design had some level of effectiveness, but it also suggested that there was a limiting factor in capturing the target insects. The limiting factor could have been a design flaw, such as the light not being attractive enough to the target insects (wasps and hornets), or it could have been due to other factors, such as the placement of the trap or the environmental conditions.

Criteria:	Data:	Did it meet the criteria:
Assembly time under 10 minutes	Less than 30 seconds	Yes
Weight less than one pound	.88 lbs	Yes
Volume under 14 cubic inches	1’x2¾”x 2¾”	Yes
insects neutralized in one location. (Outside, inside, entryway tests). At least 10 insects	Inside: 0	No
	Outside: 4	No
	Entryway:	

The data demonstrates that the insect trap successfully meets some of our high-priority design requirements; However, it did not meet one requirement, the trap needed to catch at least 10 insects in one generalized area inside the sticky tube.. The assembly time was well within the 10-minute limit, the size and weight were within the specified constraints, and the trap effectively caught insects.